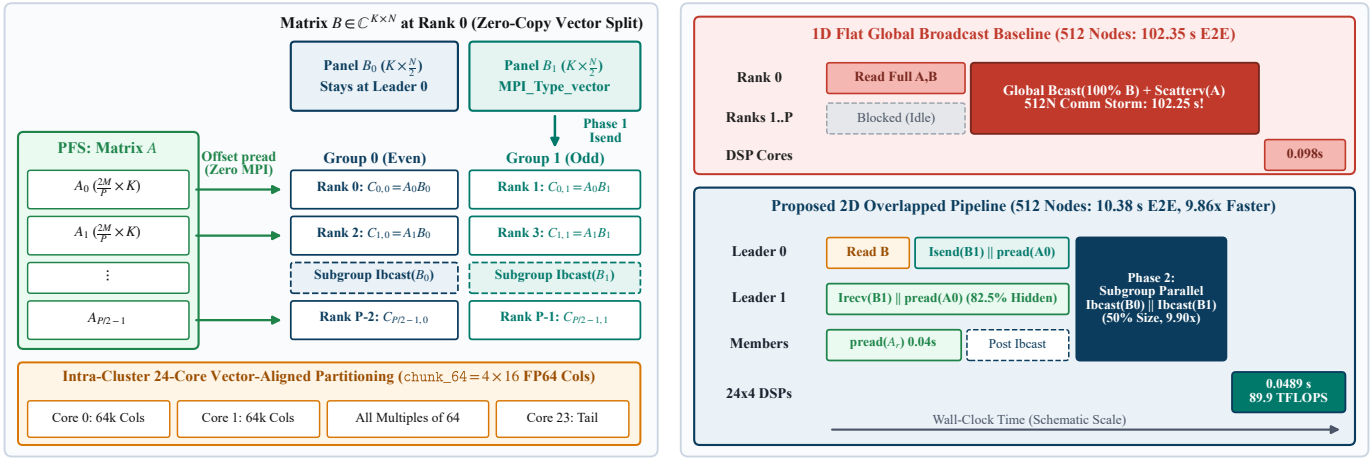

(a) Stationary-A, Panel-B 2D Grid (P/2 x 2) & Vector Alignment
Matrix $B \in \mathbb{C}^{K \times N}$ at Rank 0 (Zero-Copy Vector Split)
Panel B_0 ($K \times \frac{N}{2}$)
Stays at Leader 0
Panel B_1 ($K \times \frac{N}{2}$)
MPI_Type_vector
Phase 1
Isend
PFS: Matrix A
A_0 ($\frac{2M}{P} \times K$)
A_1 ($\frac{2M}{P} \times K$)
⋮
$A_{P/2-1}$
Offset pread
(Zero MPI)
Group 0 (Even)
Group 1 (Odd)
Rank 0: $C_{0,0} = A_0 B_0$
Rank 1: $C_{0,1} = A_0 B_1$
Rank 2: $C_{1,0} = A_1 B_0$
Rank 3: $C_{1,1} = A_1 B_1$
Subgroup Ibcast(B_0)
Subgroup Ibcast(B_1)
Rank P-2: $C_{P/2-1,0}$
Rank P-1: $C_{P/2-1,1}$
Intra-Cluster 24-Core Vector-Aligned Partitioning (chunk_64 = 4 × 16 FP64 Cols)
Core 0: 64k Cols
Core 1: 64k Cols
All Multiples of 64
Core 23: Tail
(b) 1D Global Broadcast vs. 2D Overlapped Execution Timeline
1D Flat Global Broadcast Baseline (512 Nodes: 102.35 s E2E)
Rank 0
Read Full A,B
Global Bcast(100% B) + Scatterv(A)
512N Comm Storm: 102.25 s!
Ranks 1..P
Blocked (Idle)
DSP Cores
0.098s
Proposed 2D Overlapped Pipeline (512 Nodes: 10.38 s E2E, 9.86x Faster)
Leader 0
Read B
Isend(B1) || pread(A0)
Leader 1
Irecv(B1) || pread(A0) (82.5% Hidden)
Members
pread(A_r) 0.04s
Post Ibcast
Phase 2:
Subgroup Parallel
Ibcast(B0) || Ibcast(B1)
(50% Size, 9.90x)
24x4 DSPs
0.0489 s
89.9 TFLOPS
Wall-Clock Time (Schematic Scale)